

Using Portable Video Modeling Technology to Increase the Compliment Behaviors of Children with Autism During Athletic Group Play

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Abstract A multiple baseline design across participants was used to examine the effects of a portable video modeling intervention delivered in the natural environment on the verbal compliments and compliment gestures demonstrated by five children with autism. Participants were observed playing kickball with peers and adults. In baseline, participants demonstrated few compliment behaviors. During intervention, an iPad® was used to implement the video modeling treatment during the course of the athletic game. Viewing the video rapidly increased the verbal compliments participants gave to peers. Participants also demonstrated more response variation after watching the videos. Some generalization to an untrained activity occurred and compliment gestures also occurred. Results are discussed in terms of contributions to the literature.

Keywords Autism · Video modeling · Social skills interventions · Technology

Introduction

Children across the autism spectrum often fail to demonstrate the skills required to successfully initiate and sustain

interactions (American Psychiatric Association 2013; Krasny et al. 2003; Rao et al. 2008). Commonly observed deficits include a lack of social initiations and responses, interactive play, social speech, empathy, and joint attention (Clifford and Dissanayake 2008; Rao et al. 2008; Schreibman 1988; Weiss and Harris 2001). The resulting lack of social engagement is believed to negatively impact the development of communicative, academic, adaptive, and other skills (DiSalvo and Oswald 2002; Koegel and Frea 1993; Krasny et al. 2003; Smith et al. 2002). These widespread effects highlight the importance of intervening to facilitate the acquisition of skills that promote social interaction.

One such skill may be the delivery of verbal compliments, or positively worded statements that are meant to convey approval. Compliments are common across settings and it has been suggested that giving compliments increases a person's likability (Cialdini 1993; Knapp et al. 1984). A number of treatment programs intended to increase the social competence of children with autism suggest compliment behaviors as a target (e.g., Kamps et al. 1992; Reddy 2012), but few studies have implemented interventions targeting compliment giving and directly measured changes in this behavior. One study found that prompting, reinforcement, and time out procedures effectively increased the number of compliments two children with autism and intellectual disability and one child with Down syndrome and mental retardation gave a therapist during a simple interactive ball game (Coe et al. 1990). Webb et al. (2004) implemented a social skills program for adolescents with high functioning autism that targeted several social skills including compliments. After finishing the intervention, the group of boys demonstrated statistically significant improvement in their compliment giving behavior.

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Video modeling has also been used to target compliments. Video modeling is a common intervention in which the child watches short, filmed clips of a model demonstrating the targeted behavior or behaviors and is then given the opportunity to engage in the targeted skill or skills (Bellini and Akullian 2007). This cycle is repeated until the child consistently demonstrates the targeted behaviors. To date, there is a large body of literature demonstrating that video modeling procedures effectively increase a range of social skills, including conversational speech (Charlop and Milstein 1989; Sherer et al. 2001), variation in conversational speech (Charlop et al. 2008), spontaneous verbal requests (Wert and Niesworth 2003), functional mands (Plavnick and Ferren 2011), play (MacDonald et al. 2005), and daily living skills (Shipley-Benamou et al. 2002).

Only one publication composed of two studies has used video modeling to target the compliments given by children with autism (Apple et al. 2005). In the first study, video modeling clips followed by video instruction were used to target the compliments two children with autism gave to respond to or make social initiations. After watching the video treatment package, the children increased the compliments given in response to peer initiations. However, the number of compliments given to initiate interactions did not increase until the video treatment package was combined with reinforcement procedures. In the second study, two children who watched video treatment clips again demonstrated more compliments when responding to peers but again did not use compliments to initiate interactions until additional intervention procedures were introduced. Experimentally rigorous research conducted with larger samples is needed to better examine the extent to which video modeling clips alone increases children's use of compliments.

Moreover, this past video modeling research on compliments focused solely on spoken compliments. Research has not examined the extent to which the intervention also affects the gestures that can accompany these verbalizations (Apple et al. 2005). For example, when saying "Good job," the speaker may also use a thumbs up to communicate approval. Existing research demonstrates that video modeling can effectively and simultaneously target both verbal and nonverbal behaviors (Charlop et al. 2010; Gena et al. 2005).

Additionally, few studies examine the extent to which children who participate in the interventions demonstrate response variation (i.e., exhibit novel behaviors that are part of the functional response class but were not portrayed in the modeling videos). Examining response variation is worthwhile as it provides an indication of whether the treatment has simply resulted in rote memorization of seen behaviors or facilitated children's understanding and

appropriate application of the targeted response class. Existing research indicates that video modeling interventions promote response variation. For example, Charlop and Milstein (1989) found that children who participated in a video modeling intervention designed to increase their conversational speech demonstrated more response variation during and after treatment. A later study demonstrated that a video modeling treatment consisting of multiple exemplars of conversational speech successfully increased the response variation demonstrated by two boys with autism across conversational topics (Charlop et al. 2008). More research is needed to extend these findings to portable video modeling interventions delivered in the natural environment and additional target behaviors.

The present study contributes to the literature by: (1) targeting the verbal and gestural compliments of children with autism using a portable video modeling intervention delivered in the natural environment, during the course of an athletic game, (2) assessing response variation of the verbal compliment behaviors, and (3) measuring generalization.

Method

Participants

Participants were five elementary and middle school children enrolled in an after school behavioral treatment program. All of the children were diagnosed with autism spectrum disorders, by an independent agency according to DSM-5 criteria (American Psychiatric Association 2013). According to therapist and parent report, the children demonstrated little or no complimentary behaviors towards peers (see Table 1).

Jason was 11 years, 2 months old at the start of baseline. Jason had a Vineland Adaptive Behavior Scale—Second

Table 1 Child characteristics

Child	Chron. age	Vineland-II adaptive level	Vineland-II comm. and soc. percentile
Jason	11–2	Low	1 4
Nick	9–5	Mod. Low	3 7
Mark	11–11	Adequate	42 7
Landon	10–3	Low	1 2
Kelly	10–10	–	– –

Edition (Sparrow et al. 2005) composite score of 69, which placed him in the low range of adaptive functioning. Jason's communication domain score indicated that he communicated better than 1 % of children his age and his socialization domain score put him in the 4th percentile. According to therapist and parent report, Jason demonstrated limited verbal skills and rarely commented on his surroundings. Observed speech was often monotonous, echolalic, or inaudible. It was reported that he rarely used gestures.

Nick was 9 years, 5 months old with a composite score of 85 on the Vineland that placed him in the moderately low range of adaptive functioning. Nick ranked in the 3rd and 7th percentiles for communication and socialization. He used spontaneous speech to communicate but tended to speak about a limited range of preferred topics. He seldom gave compliments. Nick was occasionally observed using gestures (e.g., pointing or pumping his fist into the air) but did not do so consistently.

Mark was 11 years, 11 months old and received a composite score of 87 on the Vineland. This indicated that he had adequate adaptive functioning. His obtained communication domain score put him in the 42nd percentile and he scored in the 7th percentile on the socialization domain. Mark occasionally spoke in short phrases and longer sentences, but tended not to do so unless prompted. According to parent and therapist reports, Mark rarely exhibited verbal compliments or gestures.

At the onset of the study, Landon was 10 years, 3 months old. A Vineland composite score of 65 indicated that his adaptive functioning was low and Landon's scores on the communication and socialization domains placed him in the 1st and 2nd percentiles. Landon's parents and therapists reported that he occasionally demonstrated spontaneous speech but that it was usually in short, incomplete phrases with a grunting voice and flat affect. Landon seldom gave verbal compliments or used gestures with his speech.

Kelly was 10 years, 10 months old. Vineland scores were not obtained for Kelly due to a lack of parent availability. Therapists reported that she frequently demonstrated spontaneous speech but many of her utterances were echolalic or contextually inappropriate. She rarely demonstrated verbal compliments and her mother reported that she often criticized peers (e.g., "That was bad"). Kelly exhibited gestures (e.g., pointing and clapping), but these gestures were often self-stimulatory and demonstrated at inappropriate times (e.g., when standing by herself or listening to instructions). She was rarely observed using gesture to communicate.

Materials

Videos were made of a familiar adult demonstrating three pairs of verbal compliments and compliment gestures (see Table 2 for the featured compliments and gestures). Each

Table 2 Modeled verbal compliments and compliment gestures

Verbal compliment	Gesture
"Wow, that went far!"	Thumbs-up
"That was a great kick!"	Fist pump
"That was a nice job"	Clapping

video began with the same clip of a ball rolling and then being kicked. This clip was immediately followed by the same three clips of the adult saying a compliment and then making a gesture (e.g., saying, "Wow, that went far!" and then using her hand to form a thumbs up). Consistent with past research, the adult model faced the camera and modeled the phrases and gestures at an exaggeratedly slow pace (e.g., Charlop and Milstein 1989; Charlop et al. 2010). The videos were approximately 20 s long and were identical except for the order of the exemplars, which was varied in an effort to promote response variation.

Videos were shown on an Apple iPad® 2. The tablet was 9.5 × 7.31 × .37 in. and weighed 1.44 pounds. The iPad® was chosen over comparable portable handheld devices because of its prevalence.

Setting

Baseline, treatment, and generalization sessions were conducted on a large lawn near the behavioral treatment center. Plastic bases were placed 15 feet apart from each other to form a diamond-shaped playing field. The area was populated by children with autism, neurotypical children, and adults. During intervention, videos were shown at second base.

Dependent Variables

Sessions were coded to measure the extent to which the children demonstrated verbal compliments and compliment gestures. A verbal compliment was operationalized as a positively worded statement that conveyed approval about a player's actions during the game and was said at an audible volume. Compliment gestures were defined as a nonverbal action that conveyed a message of approval.

Coding each demonstrated verbal compliment as a varied unscripted compliment, a repeated unscripted compliment, or a scripted compliment assessed response variation. A varied unscripted compliment was defined as a verbal compliment that was not portrayed in the video and that the child had not said before. A repeated unscripted compliment was a verbal compliment that was not seen in the video but that the child had already used at least once. A scripted compliment was operationalized as a verbal compliment that was included on the video.

Design and Procedure

A multiple baseline design across participants was used to evaluate the effects of the video modeling intervention. Baseline sessions were conducted to assess the children's use of verbal compliments and gestures prior to intervention. During intervention, children watched the videos and were then given the same opportunity to demonstrate the targeted behaviors. Generalization probes across activity were conducted during baseline and following intervention.

Baseline

Baseline sessions were conducted while the participants played kickball with each other, additional children with autism, neurotypical peers, and adults. Children were brought outside to the playing field and instructed to play kickball. Sessions were held when participants were on offense (i.e., taking their turns kicking the ball into the field). The game continued until each participating child had at least five opportunities to demonstrate compliment behaviors. An opportunity started when the child reached second base and continued until the child returned to second base. To maintain consistency across opportunities, the children did not get "out" and were only allowed to advance one base per kicker.

On the rare occasion that a child did not independently follow the rules of the game, therapists prompted the child to engage in the activity (e.g., if a child did not run to the next base when a teammate kicked the ball, a therapist verbally prompted him or her to run). The children were not instructed or prompted to demonstrate verbal compliments or gestures.

Intervention

During intervention, the children were again brought out to the playing field and instructed to play kickball. Intervention sessions were identical to baseline sessions except the child watched the treatment video the first time he or she reached second base. The child was instructed to turn towards the outfield while standing on the base and watch a video modeling video on an iPad®. The first time the video was introduced, the child watched it twice. During all subsequent sessions, the video was shown only once (Charlop and Milstein 1989). After watching the video, the child was instructed to turn around so that he or she was facing home plate and the therapist instructed the child to do what he or she saw in the video. The presentation of the video modeling video took approximately 30 s. While the video was being shown, the other children were instructed to demonstrate play behaviors (e.g., told to get ready to run or kick the ball) so that they were occupied. Then the

pitcher released a ball to the kicker and the game continued. The session continued until each participating child had a minimum of five opportunities to demonstrate compliment behaviors. Typically, one or two sessions occurred per week.

Once a child demonstrated a verbal compliment during at least four of the five opportunities, a second session was conducted without showing the child the video modeling video. If the child again said a verbal compliment on at least 80 % of the opportunities, the child was considered to have mastered the skill. If the child did not meet this criterion, the video modeling treatment was reintroduced. This cycle continued until the child did meet criterion.

Generalization

Generalization probes across activity were conducted during baseline and post-intervention. During these probes, children were instructed to play baseball. All other procedures were as previously described under baseline.

Scoring

Each session was video recorded for later coding. The child's first five opportunities to demonstrate the compliment behaviors were coded for the occurrence of verbal compliments and compliment gestures. An opportunity was scored correct for verbal compliments if the child demonstrated at least one verbal compliment and incorrect if the child did not. Coders also recorded the total number of compliments given during each session and each verbal compliment was coded as scripted or unscripted. The phrases featured in the videos were considered scripted compliments. All other verbalizations that met the operational definition were coded as unscripted compliments (e.g., "Awesome!" or "That was the best kick ever!"). An opportunity was coded correct for compliment gestures if the child made at least one compliment gesture and incorrect if the child did not.

A child was considered to have successfully completed the intervention when he or she demonstrated a verbal compliment on at least 80 % of opportunities during two consecutive sessions. Criterion only included one behavior so as to avoid an overly lengthy intervention that disrupted the children's natural play. Verbal compliments were selected because this behavior was considered the primary target of the intervention.

Inter-observer Agreement

A second coder independently scored approximately 33 % of all sessions for both dependent variables. Inter-observer agreement was calculated by dividing the number of

agreements by the number of agreements plus disagreements and multiplying that number by 100. Inter-observer agreement across children, behaviors, and treatment phases was 88 %.

Social Validity of Treatment Effects

Social validity of treatment effects was assessed by having 22 therapists rate each child's behavior pre- and post-treatment. The therapists were recruited from an after school behavioral treatment program and had experience with children with autism and neurotypical children. A short, 30 s clip was randomly selected from one of each child's baseline sessions and one of each child's final two treatment sessions. Each clip showed the child kicking the ball and running the bases. Raters watched the clips in a random order and were not told which were from baseline or post-treatment. After watching each clip, raters completed a 3-item questionnaire assessing their impressions of the child's social skills and likeability on a 7-point Likert scale. One-tailed Wilcoxon signed-rank tests were conducted to identify statistically significant differences in pre- and post-treatment ratings.

Procedural Integrity

Over 40 % of sessions were assessed for procedural integrity. Stratified random sampling was used to ensure the inclusion of at least 33 % of each child's baseline sessions, intervention sessions, and generalization probes. A coder reviewed selected sessions to determine whether the child was engaged in the appropriate athletic group play and that the video was only shown when and as described in the procedures section. Procedural integrity was 93 % for Jack's sessions, 89 % for Kelly's sessions, and 100 % for Nick, Mark, and Landon. A second coder independently scored approximately 33 % of selected treatment fidelity sessions. Inter-observer agreement was 100 %.

Results

The obtained data indicate that video modeling shown on and iPad® during an athletic group game increased the verbal compliments given by children with autism. After watching the video modeling clips, the children often demonstrated more than one compliment per opportunities and four of the five children demonstrated extensive response variation. The number of verbal compliments observed during generalization probes across activity increased following intervention, but this increase was inconsistent both within and across children. The children seldom made compliment gestures.

Verbal Compliments

Percent Opportunity

The children demonstrated few or no verbal compliments during baseline sessions. Implementation of the portable video modeling intervention increased the number of verbal compliments given. All of the children met criterion in 2–7 game sessions after the video modeling intervention was presented 1–5 times (see Fig. 1).

Jason did not exhibit any verbal compliments during baseline. Video modeling immediately increased his use of verbal compliments to 80 % of opportunities but his performance decreased when the video was not shown immediately before testing sessions. After 3 presentations of the video over five game sessions, Jason gave peers compliments on 100 % of opportunities across two consecutive sessions.

Nick demonstrated only one verbal compliment during five baseline sessions. Nick's use of compliments began to increase after video modeling was presented three times over three game sessions. He met criterion in seven game sessions after five presentations of the video modeling intervention. During his final two sessions, Nick exhibited compliments on 100 % of opportunities.

Mark did not give any verbal compliments during baseline. The video immediately increased his demonstration of verbal compliments. Mark met criterion after video modeling was presented four times over six game sessions and participating in a total of seven game sessions. Mark exhibited compliments during 90 % of opportunities over his last two sessions.

Landon did not give any verbal compliments during eight baseline sessions. After the first presentation of the video modeling intervention, Landon demonstrated verbal compliments on 100 % of opportunities. The video was not shown before his second session and Landon again gave compliments during all opportunities.

Kelly seldom gave compliments during baseline. The implementation of the video modeling intervention immediately increased her use of compliments to 80 %. Due to procedural error, the video continued to be shown before her next three sessions and she continued to give compliments during 80 % and 100 % of opportunities. The video was removed and she met criterion after 4 video modeling sessions and participating in five game sessions.

Frequency of Verbal Compliments

The percent of opportunities data show that the children increased their demonstration of compliments, but does not fully reflect the extent of the behavior change. After watching the video, the children often gave multiple

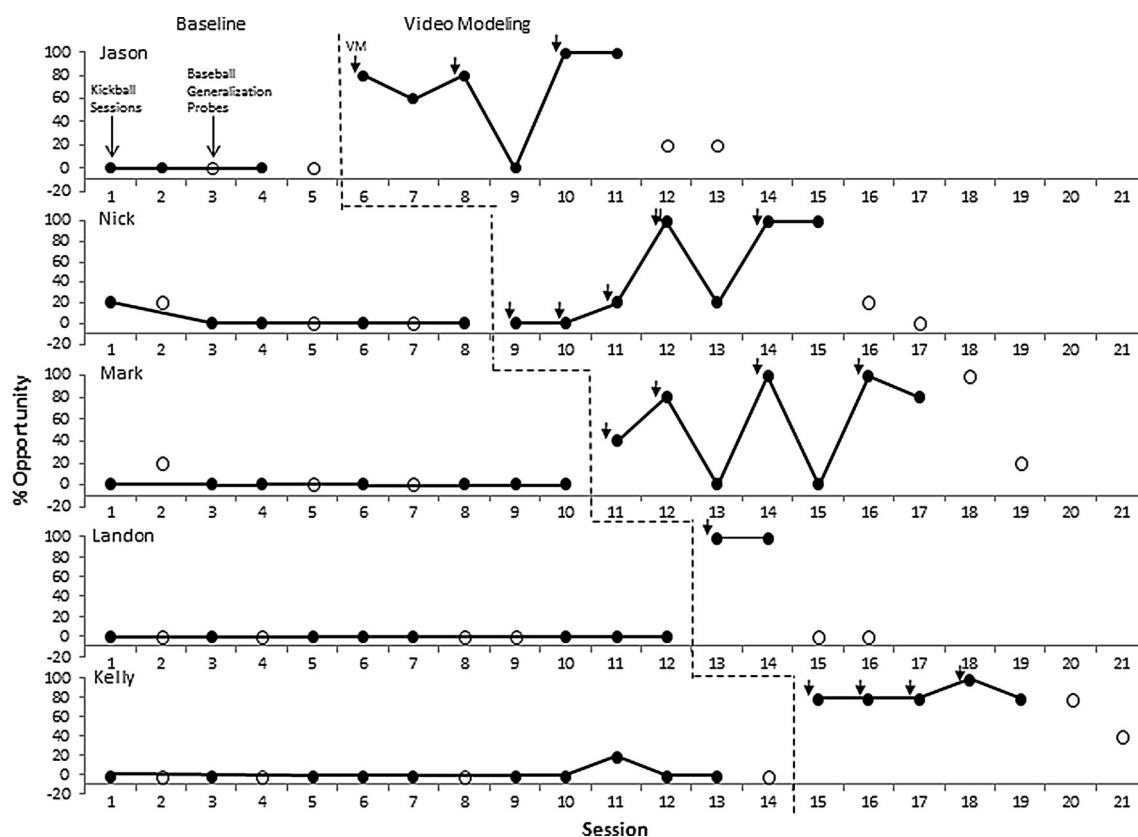


Fig. 1 Percentage of opportunities in which verbal compliments occurred during baseline, treatment, and generalization probes

compliments per opportunity and demonstrated as many as 18 verbal compliments per session. As previously reported, Jason, Mark, and Landon did not give any compliments during baseline. During the two sessions in which they met criterion, Jason demonstrated an average of six verbal compliments, Mark gave an average of 8 verbal compliments per session, and Landon said an average of 16.5 verbal compliments per session. Nick gave one verbal compliment during one interval during baseline sessions and demonstrated an average of 5.5 verbal compliments during his final two sessions. Kelly said 1 compliment across her 10 baseline sessions and averaged 10 verbal compliments during her last two intervention sessions.

Response Variation

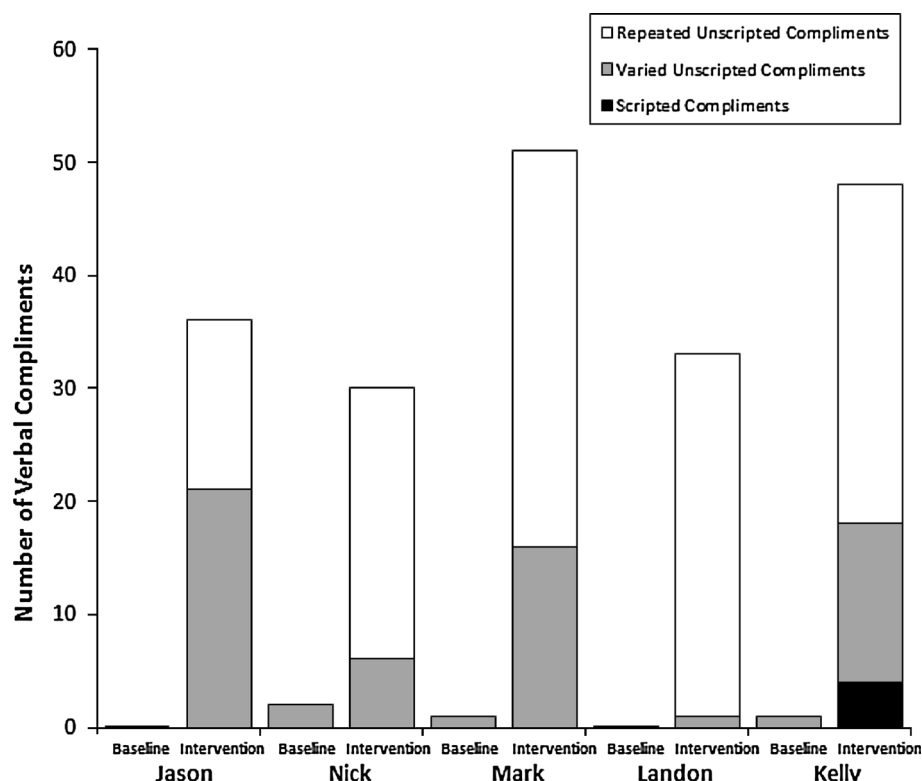
The intervention increased response variation (see Fig. 2 for the total number of varied unscripted responses, repeated unscripted responses, and scripted responses demonstrated by each participant before and after the introduction of the video modeling treatment). During baseline and baseline generalization probes, the children demonstrated 0–2 varied unscripted compliments during baseline and did

not repeat any of these compliments. After starting the video modeling treatment, all of the children increased their response variation. Four of the five children demonstrated between 6 and 21 varied unscripted compliments over their intervention sessions and post-intervention generalization probes. Landon exhibited 1 varied unscripted compliment after the onset of intervention. The majority of the other compliments given were repeated unscripted compliments. In fact, three of the children did not say any scripted compliments.

Generalization

Generalization of the observed treatment gains was limited. During baseline generalization probes, Jason, Landon, and Kelly did not demonstrate any verbal compliments and Nick and Mark each gave one compliment during one opportunity. Jason, Mark, and Kelly increased their use of compliments during baseball games following intervention, with compliments during 20, 50, and 60 % of opportunities, respectively. Nick and Landon did not increase their demonstration of compliments during post-intervention generalization probes.

Fig. 2 Cumulative number of varied unscripted, and repeated unscripted, and scripted compliments demonstrated prior to and after the introduction of the video modeling intervention across participants



Compliment Gestures

None of the children used any gestures during baseline. After the video was introduced, four of the five participants demonstrated gestures but this use was inconsistent (see Fig. 3). During his six intervention sessions, Jason demonstrated eight gestures. Nick exhibited only one gesture over his seven intervention sessions. Mark used four gestures over seven intervention sessions and Kelly demonstrated four gestures over five intervention sessions. Landon did not use any gestures during his two intervention sessions.

Social Validity of Treatment Effects

Analyses of social validity ratings indicate that therapists noted statistically significant improvements in three of the children's behavior after their participation in the portable video modeling treatment (see Table 3). Following intervention, Jason and Mark's scores were significantly higher on all three items. Landon was rated higher on two of the three items. Ratings of Nick and Kelly's behavior were not significantly higher following intervention.

Discussion

The results of this study support the use of portable video modeling interventions implemented in children's natural

environment. All participants increased their use of verbal compliments and 3 participants exhibited gains in generalization probes across activity. Four of the five participants demonstrated an increased number of compliment gestures during intervention sessions, but gains were inconsistent and did not generalize across activity.

The introduction of the video modeling treatment quickly increased the verbal compliments given by children with autism. All five children reached mastery criterion after 1–5 video modeling sessions. This rapid rate of acquisition is consistent with past video modeling research targeting other social communicative skills (e.g., Charlop et al. 2010; Charlop and Milstein 1989) but the present positive results contradicts the previous finding that a video treatment package including video modeling did not increase the compliments children with autism gave as initiations (Apple et al. 2005). These contradictory outcomes may be due to differences in the utilized video modeling procedures, the setting in which the children were expected to demonstrate the compliments, or participant characteristics. Regardless, the present study demonstrates that video modeling procedures can affect children's use of compliments as social initiations. Moreover, children in the present study demonstrated the compliments while playing an interactive athletic group game with other children. The use of this socially appropriate behavior during play is significant because many children with autism tend to withdraw and spend their time alone

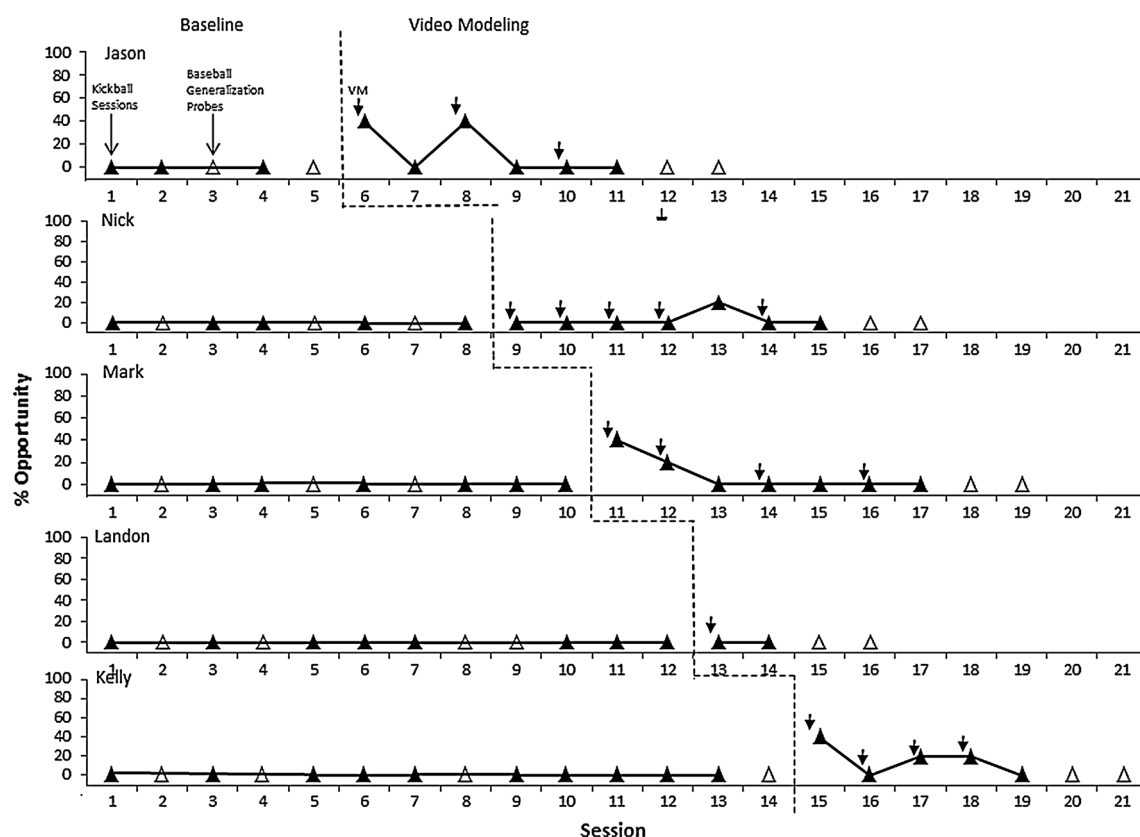


Fig. 3 Percentage of opportunities in which compliment gestures occurred during baseline, treatment, and generalization probes

Table 3 Social validity results

Item	Jason	Nick	Mark	Landon	Kelly
1. This child says nice things to other children	$p < .001$	n/s	$p = .010$	$p = .002$	n/s
2. This child is nice to play with	$p < .001$	n/s	$p = .031$	n/s	n/s
3. This child has good social skills	$p < .001$	n/s	$p = .009$	n/s	n/s

even when in close proximity to peers (Anderson et al. 2004; Lang et al. 2011).

As seen in past research, all of the children in the present study demonstrated increased response variation after starting the intervention (Charlop and Milstein 1989; Charlop et al. 2008). Examples of these varied unscripted compliment include “Nice slide!”, “Awesome!”, and “That was a big kabloomie whammo!”. These findings are significant as they indicate that the children were not simply memorizing scripted phrases. Instead, the children watched the videos and adapted the information to their current environment. This may have resulted at least in part

from the inclusion of multiple exemplars in the treatment videos (e.g., Charlop et al. 2008; Stokes and Baer 1977).

This study is one of the first to demonstrate that portable video modeling interventions implemented during naturally occurring social events can effectively target at least certain skills of children with autism. Past studies of the use of video modeling on portable technology have included additional prompting procedures (Cardon 2012; Cihak et al. 2010) or lacked the experimental control needed to attribute observed gains to the video modeling treatment (Kagohara et al. 2012). It is worth noting that the children in the present study demonstrated the gains after watching the videos while participating in the athletic activity. The finding that the observed increase occurred after the children watched the video in a rather chaotic and distracting environment containing numerous peers and others suggests that portable video modeling interventions may effectively target skills in similar school or community settings. The possibility of targeting skills within children’s daily routines is important because children with autism who make gains in one setting often fail to apply these skills when in different settings or with different people and stimuli (Schreibman 2000; Stokes and Baer 1977).

The social validity data indicate that the observed increases in the children's compliment-giving behaviors improved therapists' ratings of Jason, Mark, and Landon's social behavior. However, therapists did not find Nick and Kelly's behavior to be significantly better after intervention. This pattern of findings may be related to the children's demonstration of other social behaviors. Standardized test scores and clinical observations indicate that Nick and Kelly initially demonstrated more verbal and interactive behaviors than the other participants. The prevalence of these other social behaviors may make the children's increased demonstration of compliments less noticeable or significant to the therapists.

While the stimuli, activities, and targeted compliments were common in participating children's typical routines and the videos portrayed multiple exemplars of the targeted behaviors (Goldstein 2002; Stokes and Baer 1977), generalization was not pervasive. Although all children acquired verbal compliments, it is possible that fluency in the verbal compliment behavior did not occur. Fluency refers to the ability to consistently demonstrate skills with speed and accuracy (Charlop and Carpenter 2000; Kubina and Wolfe 2005). Fluent responding increases the likelihood that demonstration of the behavior will activate natural contingencies and is thought to promote generalization. Achieving fluency often requires numerous learning opportunities (Green 2001). In the present study, the children met the acquisition criterion so quickly that they may not have practiced the skill to fluency. It is possible that extending the length of intervention to provide children with more learning opportunities would allow the children to become fluent in the behavior and better facilitate generalization. Kelly's data provides preliminary support for this modification. Kelly continued to view the video and participate in sessions after consistently demonstrating verbal compliments and also gave the most compliments during generalization probes.

Another limitation of the present study is the children's minimal increases in compliment gestures. It is important to note that the children were not required to meet a learning criterion for gestures. This was because verbal compliments were considered the primary target to avoid an overly lengthy intervention that disrupted the children's play. While past research indicates that video modeling can effectively target both verbal and nonverbal behaviors simultaneously (Charlop et al. 2010; Gena et al. 2005), these prior studies did not intervene in the natural environment in the middle of an ongoing athletic game. Further, data indicate that the children tended to acquire the targeted verbalizations more rapidly than the gestures (Charlop et al. 2010). Additional presentations of the video modeling treatment may have facilitated children's use of gestures. Also, while both verbal compliments and compliment

gestures were displayed in the videos, the verbal compliment was always presented first. It is possible that this made the verbal compliments more salient to the children and that they identified this as the only target behavior. Another possibility is that the peers responded differently to demonstrated verbal and gesture compliments and that this contributed to the observed difference in frequency. While data was not collected on peers' responses to verbal or gestural compliments, it was anecdotally observed that the children often responded to verbal compliments by saying "Thanks!" or "Yeah!". These natural verbal responses may have differentially reinforced the verbal complimenting behaviors and inadvertently discouraged children's use of the modeled gestures. Additionally, there is a need for empirical research identifying the extent to which neurotypical children demonstrate compliments during athletic activities. This data would be useful in developing a socially valid mastery criteria.

The present study demonstrates that video modeling on portable handheld technology in naturalistic environments can effectively increase social skills of children with autism. Watching the video models on an iPad® during a kickball game rapidly increased children's use of untrained and varied compliments. While more research is needed, this finding has important implications for interventions for children with autism. Portable video modeling interventions may be a relatively easy to implement and efficient means of directly affecting social behaviors within children's behavior in natural settings.

Conflict of interest The authors declare that they have no conflict of interest.

References

- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Arlington, VA: American Psychiatric Publishing.
- Anderson, A., Moore, D. W., Godfrey, R., & Fletcher-Flinn, C. M. (2004). Social skills assessment of children with autism in free-play situations. *Autism*, 8(4), 369–385. doi:10.1177/1362361304045216.
- Apple, A. L., Billingsley, F., & Schwartz, I. S. (2005). Effects of video modeling alone and with self-management on compliment-giving behaviors of children with high-functioning ASD. *Journal of Positive Behavior Interventions*, 7(1), 33–46.
- Bellini, S., & Akullian, J. (2007). A meta-analysis of video modeling and video self-modeling interventions for children and adolescents with autism spectrum disorders. *Exceptional Children*, 73, 264–287.
- Cardon, T. A. (2012). Teaching caregivers to implement video modeling imitation training via iPad for their children with autism. *Research in Autism Spectrum Disorders*, 6(4), 1389–1400. doi:10.1016/j.rasd.2012.06.002.
- Charlop, M. H., & Carpenter, M. H. (2000). Modified incidental teaching sessions: A procedure for parents to increase spontaneous speech in their children with autism. *Journal of Positive*

- Behavior Interventions*, 2(2), 98–112. doi:10.1177/109830070000200203.
- Charlop, M. H., Dennis, B., Carpenter, M. H., & Greenberg, A. L. (2010). Teaching socially expressive behaviors to children with autism through video modeling. *Education and Treatment of Children*, 33(3), 371–393. doi:10.1353/etc.0.0104.
- Charlop, M. H., Gilmore, L., & Chang, G. T. (2008). Using video modeling to increase variation in the conversation of children with autism. *Journal of Special Education Technology*, 23(3), 47–66.
- Charlop, M. H., & Milstein, J. P. (1989). Teaching autistic children conversational speech using video modeling. *Journal of Applied Behavior Analysis*, 22, 275–285. doi:10.1901/jaba.1989.22-275.
- Cialdini, R. B. (1993). *Influence: Science and practice* (3rd ed.). New York: Harper Collins College Publishers.
- Cihak, D., Fahrenkrog, C., Ayres, K. M., & Smith, C. (2010). The use of video modeling via a video iPod and a system of least prompts to improve transitional behaviors for students with autism spectrum disorders in the general education classroom. *Journal of Positive Behavior Interventions*, 12, 103–115. doi:10.1177/1098300709332346.
- Clifford, S. M., & Dissanayake, C. (2008). Early development of joint attention in infants with autistic disorder using home video observations and parental interview. *Journal of Autism and Developmental Disorders*, 38, 791–805. doi:10.1007/s10803-007-0444-7.
- Coe, D., Matson, J., Fee, V., Manikam, R., & Linarello, C. (1990). Training nonverbal and verbal play skills to mentally retarded and autistic children. *Journal of Autism and Developmental Disorders*, 20, 177–187. doi:10.1007/BF02284717.
- DiSalvo, C. A., & Oswald, D. P. (2002). Peer-mediated interventions to increase the social interaction of children with autism: Consideration of peer expectancies. *Focus on Autism and Other Developmental Disabilities*, 17(4), 198–207. doi:10.1177/10883576020170040201.
- Gena, A., Couloura, S., & Kymissis, E. (2005). Modifying the affective behavior of preschoolers with autism using in vivo or video modeling and reinforcement contingencies. *Journal of Autism and Developmental Disorders*, 35(5), 545–555. doi:10.1007/s10803-005-0014-9.
- Goldstein, H. (2002). Communication intervention for children with autism: A review of treatment efficacy. *Journal of Autism and Developmental Disorders*, 32(5), 373–396. doi:10.1023/A:1020589821992.
- Green, G. (2001). Behavior analytic instruction for learners with autism: Advances in stimulus control technology. *Focus on Autism and Other Developmental Disabilities*, 16(2), 72–85. doi:10.1177/108835760101600203.
- Kagohara, D. M., Sigafoos, J., Achmadi, D., O'Reilly, M., & Lancioni, G. (2012). Teaching children with autism spectrum disorders to check the spelling of words. *Research on Autism Spectrum Disorders*, 6(1), 304–310. doi:10.1016/j.rasd.2011.05.012.
- Kamps, D. M., Leonard, B. R., Vernon, S., Dugan, P., & Delquadri, J. C. (1992). Teaching social skills to students with autism to increase peer interactions in an integrated first-grade classroom. *Journal of Applied Behavior Analysis*, 25(2), 281–288.
- Knapp, M. L., Hopper, R., & Bell, R. A. (1984). Compliments: A descriptive taxonomy. *Journal of Communication*, 34(4), 12–31. doi:10.1111/j.1460-2466.1984.tb02185.x.
- Koegel, R. L., & Frea, W. D. (1993). Treatment of social behavior in autism through the modification of pivotal social skills. *Journal of Applied Behavior Analysis*, 26(3), 369–377. doi:10.1901/jaba.1993.26-369.
- Krasny, L., Williams, B., Provencal, S., & Ozonoff, S. (2003). Social skills interventions for the autism spectrum: Essential ingredients and a model curriculum. *Child and Adolescent Psychiatric Clinics of North America*, 12, 107–122. doi:10.1016/S1056-4993(02)00051-2.
- Kubina, R. M., & Wolfe, P. (2005). Potential applications of behavioral fluency for students with autism. *Exceptionality: A Special Education Journal*, 13(1), 35–44. doi:10.1207/s15327035ex1301_5.
- Lang, R., Kuriakose, S., Lyons, G., Mulloy, A., Boutot, A., Britt, C., et al. (2011). Use of school recess time in the education and treatment of children with autism spectrum disorders: A systematic review. *Research in Autism Spectrum Disorders*, 5(4), 1296–1305. doi:10.1016/j.rasd.2011.02.012.
- MacDonald, R., Clark, M., Garrigan, E., & Vangala, M. (2005). Using video modeling to teach pretend play to children with autism. *Journal of Behavioral Interventions*, 20, 225–238. doi:10.1002/bin.197.
- Plavnick, J. B., & Ferren, S. J. (2011). Establishing verbal repertoires in children with autism using function-based video modeling. *Journal of Applied Behavior Analysis*, 44(4), 747–766. doi:10.1901/jaba.2011.44-747.
- Rao, P. A., Beidel, D. C., & Murray, M. J. (2008). Social skills interventions for children with Asperger's Syndrome or high-functioning autism: A review and recommendations. *Journal of Autism and Developmental Disorders*, 38(2), 353–361. doi:10.1007/s10803-007-0402-4.
- Reddy, L. A. (2012). *Group play interventions for children: Strategies for teaching prosocial skills*. Washington, D.C.: APA Books.
- Schreibman, L. (1988). *Autism*. Newbury Park, CA: Sage Publications.
- Schreibman, L. (2000). Intensive behavioral/psychoeducational treatments for autism: Research needs and future directions. *Journal of Autism and Developmental Disorders*, 30, 373–378. doi:10.1023/A:1005535120023.
- Sherer, M., Pierce, K. L., Paredes, S., Kisacky, K. L., Ingersoll, B., & Schreibman, L. (2001). Enhancing conversation skills in children with autism via video technology: Which is better, 'Self' or 'Other' as a model? *Behavior Modification*, 25(1), 140–158. doi:10.1177/0145445501251008.
- Shipley-Benamou, R., Lutzker, J. R., & Taubman, M. (2002). Teaching daily living skills to children with autism through instructional video modeling. *Journal of Positive Behavior Interventions*, 4, 165–175.
- Smith, T., Lovaas, N. W., & Lovaas, O. I. (2002). Behaviors of children with high-functioning autism when paired with typically developing versus delayed peers: A preliminary study. *Behavioral Interventions*, 17(3), 129–143. doi:10.1002/bin.114.
- Sparrow, S., Cicchetti, D., & Balla, D. (2005). *Vineland-II: Vineland adaptive behavior scales: Survey forms manual* (2nd ed.). Circle Pines, MN: American Guidance Services.
- Stokes, T., & Baer, D. (1977). An implicit technology of generalization. *Journal of Applied Behavior Analysis*, 10, 349–367. doi:10.1901/jaba.1977.10-349.
- Webb, B. J., Miller, S. P., Pierce, T. B., Strawser, S., & Jones, W. P. (2004). Effects of social skill instruction for high-functioning adolescents with autism spectrum disorders. *Focus on Autism and Developmental Disabilities*, 19(1), 53–62. doi:10.1177/10883576040190010701.
- Weiss, M. J., & Harris, S. L. (2001). Teaching social skills to people with autism. *Behavior Modification*, 25(5), 785–802. doi:10.1177/0145445501255007.
- Wert, B. Y., & Niesworth, J. T. (2003). Effects of video self-modeling on spontaneous requesting in children with autism. *Journal of Positive Behavior Interventions*, 5(1), 30–34. doi:10.1177/10983007030050010501.